

William Cook

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EDUCATION

Liberty University

Expected May 2028

B.S. Computer Science: Software Engineering & Technology Entrepreneurship

Lynchburg, VA

- **GPA:** 3.78/4.0 | **Coursework:** Data Structures & Algorithms, Operating Systems Design, Computer Architecture, Database Design, Information Security, Network Architecture & Protocols, Advanced C++

EXPERIENCE

Cole Design & Development LLC

December 2025 – Present

Software Engineering Intern

Hudson, OH

ColeCloud (IoT device management platform)

July 2026

- Built an IoT device management platform (Node.js, Express, PostgreSQL, MQTT, React, Docker) being prepared for rollout to the company's clients. Sole developer; live on AWS EC2 in three weeks.
- Modeled device capabilities as declarative JSON and telemetry as JSONB, so no SQL or business logic names a specific metric. Four device types run on it; adding another takes only a config entry.
- Set up per-device X.509 certificates and mutual TLS on the MQTT broker, with a CA that signs each device cert at provisioning time, so a stolen credential only affects the one device that leaked it. Found and closed an anonymous MQTT-over-WebSockets listener exposed to the internet.
- Wrote a 32-endpoint REST API, WebSocket push layer, MQTT ingestion pipeline, and server-side alerting engine behind an eight-view React dashboard.
- Tracked a production outage to a full EC2 root volume (uncapped Docker build cache) that killed Postgres mid-init. Fixed two event-loop blockers: a synchronous OpenSSL spawn and an awaited webhook.

Embedded and desktop tooling

Dec 2025 – Jun 2026

- Added Rust/Tauri backend commands that expose an embedded device's protocol layer to a TypeScript/React frontend, so the UI could finally drive the hardware directly.
- Closed GitHub issues on config persistence, frontend/backend integration, and parameter locking. Mapped an embedded C module's external function and global dependencies ahead of a refactor.

Missionize

June 2025 – September 2025

Full Stack Developer

Remote

- Built React Native screens for habit tracking, journaling, and dashboards on Discipler.io, an AI habit-tracking app for church ministries.
- Wrote FastAPI endpoints against MongoDB and worked through feature branches and code review.

Liberty University Competitive Programming Club

August 2024 – Present

Officer

Lynchburg, VA

- Took a team to 6th of 61 at the 2024 ICPC Mid-Atlantic Regional. Solved 150+ Kattis problems and teach weekly C++ and Python algorithm sessions.

PROJECTS

Self-Hosted Server Infrastructure | *Linux, Raspberry Pi 5, Python, Nginx, systemd, Tailscale*

Ongoing

- Run a headless Linux server (Raspberry Pi 5, 16GB RAM, 1TB NVMe) hosting several services, with SSH key auth, Tailscale for remote access, Nginx and HTTPS, and cron backups.
- Wrote an async Python service that tails logs and forwards events off-box, running as a systemd unit.

TECHNICAL SKILLS

Languages: Python, C++, C (Embedded), Rust, TypeScript, JavaScript, SQL, Bash, HTML/CSS

Frameworks: React, React Native, Node.js, Express, FastAPI, Tauri, Vite, Tkinter, paho-mqtt

Infrastructure & Data: Docker, AWS (EC2, S3), Azure, PostgreSQL, MongoDB, Linux, Nginx, Caddy, systemd, Git

Protocols & Systems: MQTT, WebSockets, REST API design, mutual TLS & X.509 PKI, embedded systems, Linux admin